

9. Modal Summary and Harmonic Response

The harmonic response analysis uses the mode-superposition method with a linear frequency sweep from 1 Hz to 150 Hz over 150 solution intervals. Four harmonic 32 N forces act in the +Z direction at the motor mounting locations.

Setting	Value
Method	Mode Superposition
Frequency spacing	Linear
Frequency range	1–150 Hz
Intervals	150
Excitation	32 N per arm, +Z

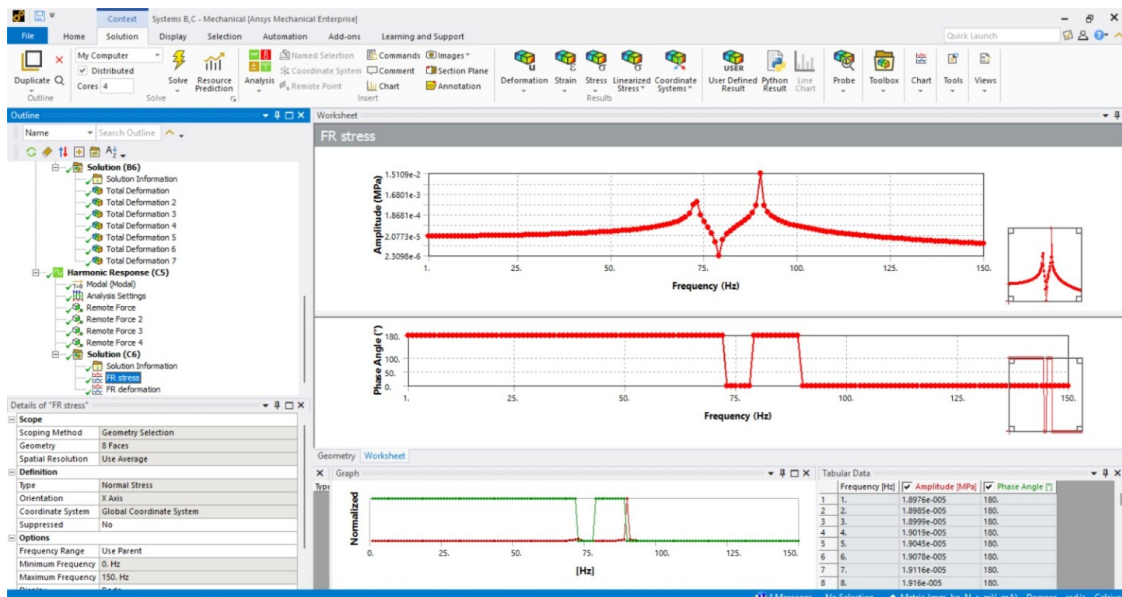


Figure. Harmonic normal-stress amplitude and phase versus frequency.

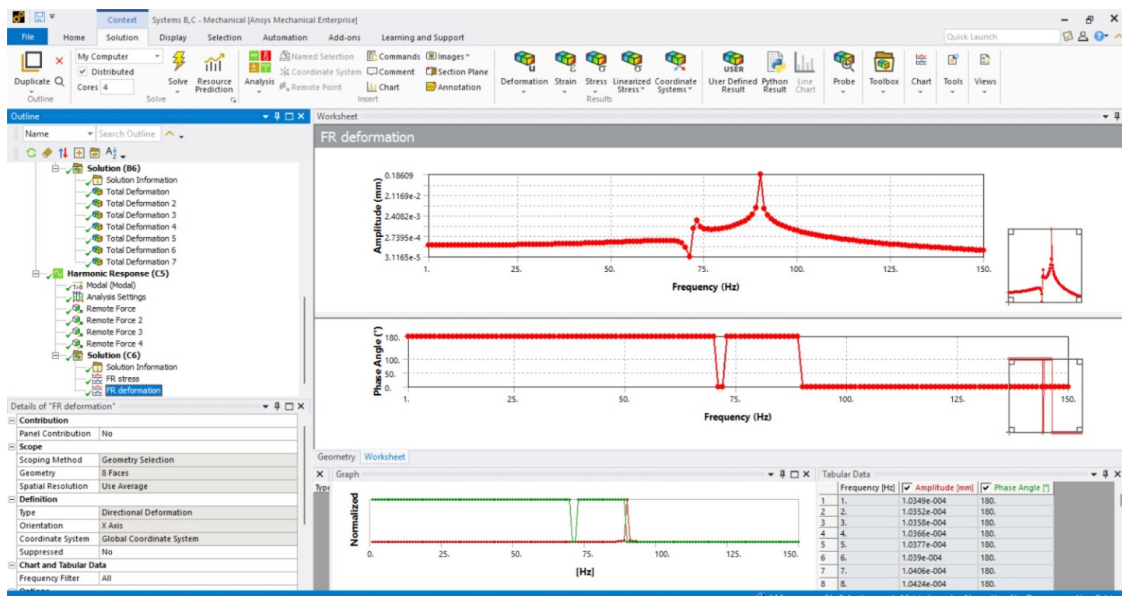


Figure. Harmonic deformation amplitude and phase versus frequency.

10. Harmonic Results and Resonance

The harmonic response remains comparatively low through most of the frequency sweep, but a pronounced response peak appears near 90 Hz. The supplied response plot indicates an approximate maximum normal-stress amplitude of 0.0151 MPa and deformation amplitude of approximately 0.186 mm in this resonance region.

The resonance peak closely coincides with Modes 5 and 6 at 89.963 Hz and 89.971 Hz. This agreement between the independent modal frequency extraction and the harmonic frequency sweep is the strongest dynamic finding of the study.

Dynamic quantity	Observed value
First modal cluster	72.582–72.620 Hz
Second modal cluster	89.963–89.971 Hz
Harmonic peak	≈90 Hz
Peak normal-stress amplitude	≈0.0151 MPa
Peak deformation amplitude	≈0.186 mm

From a design perspective, the ≈90 Hz region should be treated as the principal vibration-sensitive band identified by this simulation. The final motor/propeller operating frequencies and their dominant harmonics should be checked against the natural frequencies before continuous operation in this region.